
Stefanos Fragkoulis, Ph.D.

Athens, Greece

(+30) 697 614 2310 (WhatsApp/Viber)
stefanos_fragkoulis@hotmail.com
[LinkedIn](#) | [GitHub](#) | [ORCID](#)

Professional Summary

Bioinformatician & Data Scientist combining a Ph.D. in Life Sciences with a recent M.Sc. in Applied Bioinformatics and formal certifications in Machine Learning and Python. Offers 10+ years of experience engineering reproducible data pipelines, executing advanced statistical modelling, and translating high-dimensional biological datasets into actionable insights. Expert in R and Python with a proven track record of coordinating complex cross-functional workflows, directing large-scale technical evaluations, and communicating analytical findings to diverse scientific stakeholders.

Skills

Programming & Environment: Python (Pandas, NumPy, Scikit-learn, Keras), R (Bioconductor, Tidyverse, data.table), Unix/Linux command-line, Bash scripting, Git/GitHub

Bioinformatics & Genomics: Systems Biology Pipelines, Network Analysis, Genome Assembly, Differential Expression (DESeq2), BEDTools

Data Science & ML: Advanced Statistical Modelling, PCA, CVA, Clustering (K-Means), Regression, Feature Extraction, Geometric Morphometrics

Data Visualization: Matplotlib, Seaborn, ggplot2, UpsetR, ComplexHeatmap

Professional Experience

Science Officer & Panel Coordinator (Life Sciences)

Hellenic Foundation for Research and Innovation (HFRI) | Athens, Greece | September 2022 - present

- **Directed the end-to-end evaluation lifecycle** for 8 high-stakes Life Sciences funding panels, overseeing portfolios of total 1017 grant proposals and funding research projects valued at over €19M
- **Synthesized complex qualitative and quantitative data** across 1017 applications, distilling multi-layered expert evaluations into actionable consensus reports for executive funding decisions
- **Coordinated cross-functional workflows** for national and international domain experts, ensuring strict data integrity, regulatory compliance, and on-time milestone delivery

Senior Data Analyst & Research Scientist

European & National Research Consortia (H2020, FP7, NSRF) | June 2012 – March 2022

- **Directed end-to-end data strategy and comparative statistical modelling** across 5 major research grants, translating complex biological data into actionable insights
- **Extracted and quantified high-dimensional traits** across thousands of samples (PerformFISH - Horizon 2020) to calculate complex genetic indices and model variance components
- **Collaborated with international stakeholders and domain experts** to synthesize complex morphometric datasets into actionable developmental models, delivering key project milestones and 12 peer-reviewed publications

Technical Projects

Systems Biology & Genome Organization Analysis Pipeline | *MSc Data Project* | June 2025 – March 2026
Tech Stack: R (Tidyverse, Bioconductor, ggplot2), Statistical Modelling, Network Analysis ([Github](#))

- **Engineered an end-to-end bioinformatics data pipeline** in R to process, clean, and analyse longitudinal genomic datasets comprising over 350,000 observations
- **Applied advanced statistical methodologies and network analysis** to uncover complex genome organization patterns, effectively reducing high-dimensional data into actionable variables

- **Designed programmatic data visualizations** using ggplot2 to translate multi-layered genomic networks into clear insights for technical presentations

Education

- **M.Sc. in Applied Bioinformatics and Data Analysis** | November 2024 – March 2026
Democritus University of Thrace, Department of Molecular Biology & Genetics, Greece
- **Ph.D. in Fish Biology** | June 2016 – December 2020
University of Crete, Biology Department, Greece
- **M.Sc. in Environmental Biology** | October 2012 – November 2014
University of Crete, Biology Department, Greece
- **B.Sc. in Biology** | October 2005 – March 2012
University of Patras, School of Science, Greece

Selected Publications & Presentations (12 Total, 8 as First Author)

- **Peer-Reviewed Literature:** *Authored 12 peer-reviewed publications (8 as first author) in high-impact international scientific journals, specializing in statistical modelling, feature extraction, and biological data analysis*
- **Conference Presentations:** *Delivered 11 oral and poster presentations at national and international scientific conferences, successfully translating complex biometric and computational concepts into clear insights for diverse audiences*

Key Journal Text Selections:

- **Fragkoulis S., Kerasovitis D., Batargias C. & Koumoundouros G. (2021)** *Body-shape trajectories and their genetic variance component in Gilthead seabream (Sparus aurata L.).* **Scientific Reports** – *Extracted complex biometric features to model quantitative genetic indices and variance components.*
- **Fragkoulis S. & Koumoundouros G. (2021)** *Simple morphometrics for predicting lordosis-induced deviations of body-shape in reared Gilthead seabream (Sparus aurata L.).* **Journal of Fish Diseases** – *Developed predictive morphological models for deformity detection, successfully translating physical traits into quantifiable predictive metrics.*
- **Fragkoulis S., Batargias C., Kolios P. & Koumoundouros G. (2019)** *Genetic parameters of the upper-jaw abnormalities in Gilthead seabream Sparus aurata.* **Aquaculture** – *Designed and executed the end-to-end data analysis for a longitudinal study on anomaly recovery and morphological adaptation.*

Certificates

- **The Complete Python Developer**, Udemy (2025)
- **Machine Learning Track**, Galaxy Training Academy (2025)
- **PRINCE2 Foundation Certificate in Project Management**, Axelos (2023)
- **Data Science: Foundations using R**, Coursera (2022)

Professional Memberships

- **International Society for Computational Biology (ISCB), Professional Member (2025 – present)**

Leadership & Additional Information

- **Volunteer Samaritan Cadet & Team Leader:** Hellenic Red Cross (Rescuers & First Aid Corps) | Oct 2025 – Present. Logged 150+ hours of field coverage; managed emergency response logistics mirroring PRINCE2 framework principles
- **Interests:** Martial Arts (Brazilian Jiu-Jitsu, Kyokushin Karate), Hiking and Outdoor Endurance Activities